



Getting Started with Geographic Information Systems (GIS)

Intro to GIS + Map Creation Basics

WORKSHOP

VANESSA CASILLAS · DANIEL PEREZ

What We'll Cover Today

Part 1 — This Session

- What GIS is and why it matters
- Basic concepts of spatial data
- How to create a simple map

Part 2 — Hands-On Session

- Building a map step-by-step
- Adding and styling data layers
- Exploring data and creating a final map product

What is GIS?

Geographic Information Systems (GIS) is a technology for capturing, storing, analyzing, and visualizing geographic data. In simple terms: **GIS connects data + location** to help us understand the world.

Capture

Collect geographic data from the real world

Store

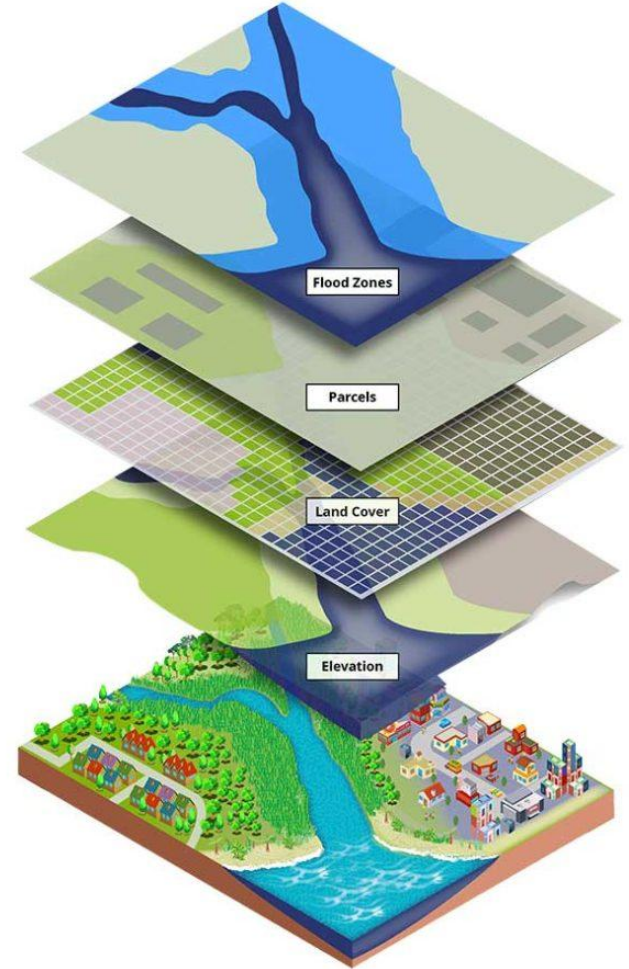
Manage and organize spatial information

Analyze

Discover patterns and relationships

Visualize

Display insights on interactive maps



What Makes GIS Powerful?

GIS helps answer critical questions — and is used across many fields.



Where are things?

Locate resources, hazards, or services precisely.



What patterns exist?

Reveal trends invisible in spreadsheets.



How do places relate?

Understand spatial relationships between locations.



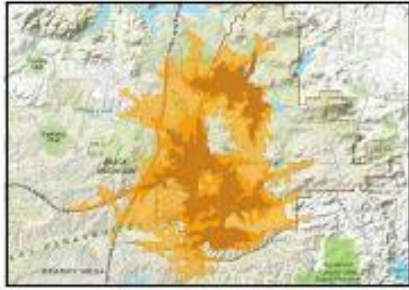
What has changed?

Track changes over time across a landscape.



Used in: Public health · Disaster response · Transportation · Equity analysis · Campus planning

Understand Places



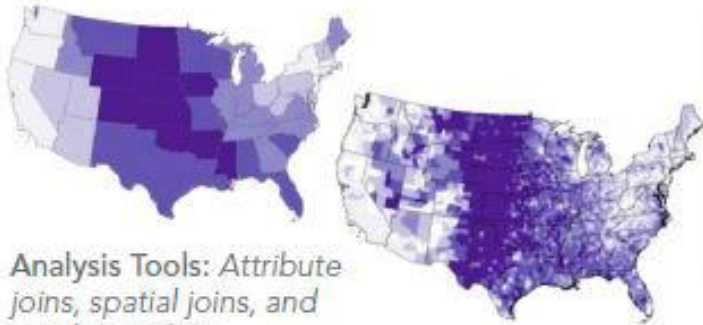
Analysis Tools: Attribute queries, spatial queries; proximity analysis

Detect Patterns



Analysis Tools: Density analysis and cluster analysis

Determine Relationships



Analysis Tools: Attribute joins, spatial joins, and overlay analysis

Make Predictions



Analysis Tools: Interpolation, regression, surface analysis

Find Locations



Analysis Tools: Site suitability, location-allocation, cost corridors



Bus and Rail System Overview map

DCR Design LLC
by DCR Design

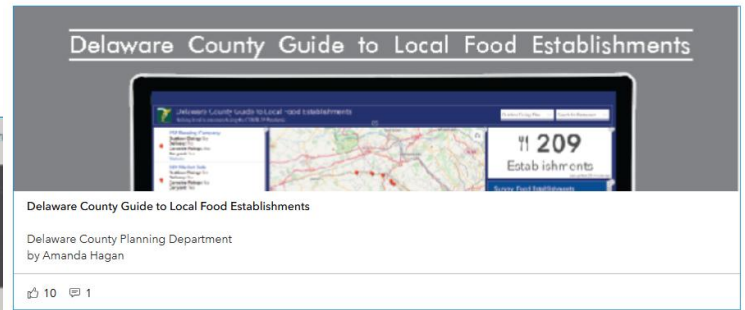
👍 14 💬 0



Nevada's Population Change: A Review of Population Growth and Decline in the State of Nevada

Legislative Counsel Bureau
by Hildi Cruz and Robin Evans

👍 8 💬 0



Delaware County Guide to Local Food Establishments

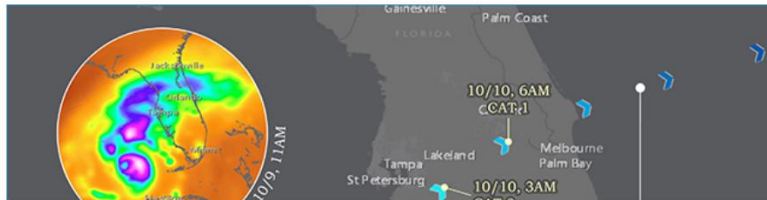
Delaware County Planning Department
by Amanda Hagen

👍 10 💬 1

<https://mapgallery.esri.com/>



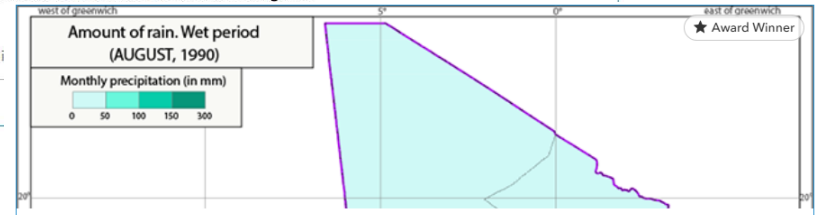
Protected Areas Database of the United States (PAD-US) - IUCN Protected Area Categories



Hurricane Milton Strength Timeline, 10/5/2024 - 10/10/2024

NASA
by Ariana L. McDermott, Lauren K. Hill-Beaton

👍 5 💬 0



Amount of Rain in Mali in Wet and Dry Periods

Africa Business and Consulting Mali
by Souleymane Sanogo

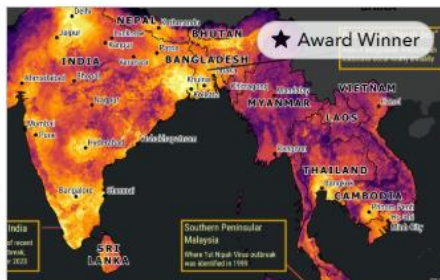
👍 47 💬 17



Willamette Valley American Viticultural Area - A Geologic History of Soils

Cascade GIS
by Erica McCormick and Rick Zeeb

👍 7 💬 0



Nipah Virus: Risk Analysis of Outbreak Potential in South and Southeast Asia

Tufts University
by Shannon O'Connor

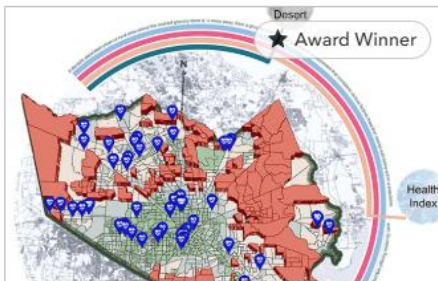
👍 10 💬 0



Intelligent Supervision with Geospatial Technology

Osinermin (Lima, null)
by ROGER LOPEZ TUESTA; LIZBETH TORRES...

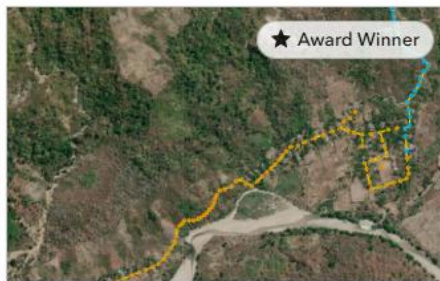
👍 578 💬 0



Geography of Maternity Care Deserts

Harris County
by Meliadis Barbara, L. Venigalla, J. Wong, A....

👍 575 💬 0



Listrik Desa (Lisdes) 2.0: Electrifying the Nation with Renewable Energy

PT PLN (Persero) Kantor Pusat
by Wahyu Ahadi Rouzi

👍 190 💬 0



Ghana Cocoa Board

Ghana Cocoa Board
by Felix Owusu Anyimah

👍 27 💬 0

Core Idea: Data + Location

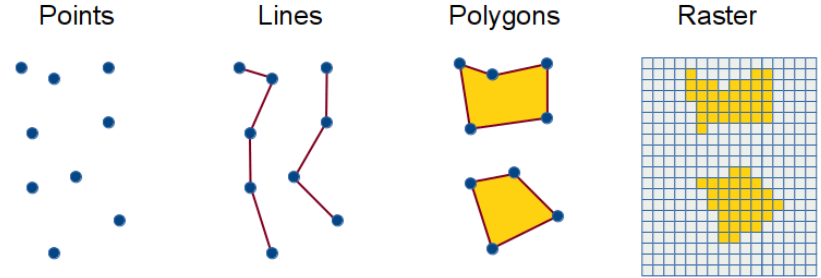
Every GIS dataset has two essential parts working together.

Spatial Data — Where

- **Points** — trees, buildings, schools
- **Lines** — roads, rivers, pipelines
- **Polygons** — boundaries, campuses, zones

Attribute Data — What

- **Name** — label or identifier
- **Type** — category or classification
- **Population** — numeric values
- **Status** — condition or state
- **Any description** — additional context



Key GIS Components



Basemap

The background reference layer — streets, terrain, or satellite imagery.



Data Layers

Your custom geographic information overlaid on the basemap.



Symbology

Colors, shapes, and sizes that represent different data values.



Pop-ups

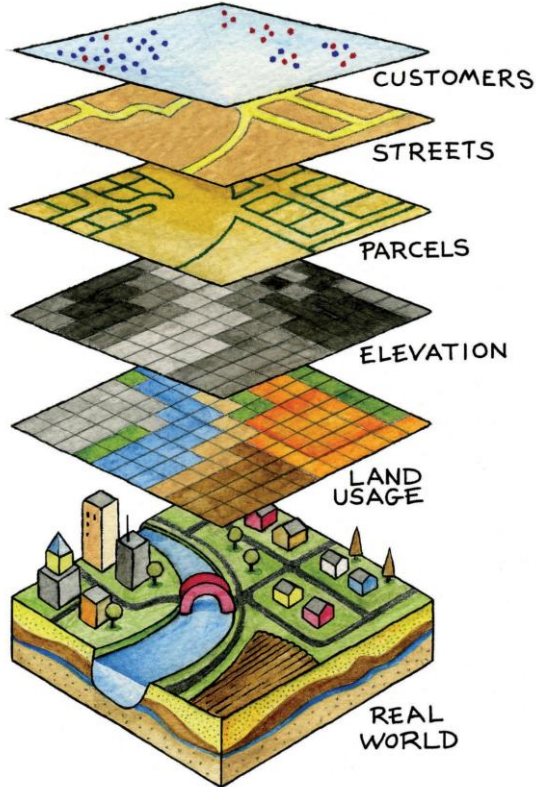
Information windows that appear when you click a feature.



Legend

A guide explaining what each symbol and color means.

What is a Map in GIS?



A GIS map is **more than a static image**. It's a dynamic collection of **layers** — each holding a different dataset that can be turned on or off, styled, and analyzed.

- Think of it like a stack of transparent sheets. Each sheet represents a different dataset — roads, buildings, parks — layered together to tell a complete story.

How to Make a Map: Basic Workflow

01

Define Your Question

What are you trying to show? Who is the audience?

02

Find or Load Your Data

CSV files, GIS servers, open data portals, or institutional datasets.

03

Add a Basemap

Choose a background reference map.

04

Add Data Layers

Load your spatial data as points, lines, or polygons.

05

Style Your Layers

Apply colors, symbols, and sizes to represent your data.

06

Configure Pop-ups

Set up information windows for map features.

07

Review and Interpret

Check your map and draw conclusions from what you see.

Step 1: Define Your Purpose



Before building any map, start with a clear purpose.

Ask yourself:

What am I trying to show?

Define the core message or insight.

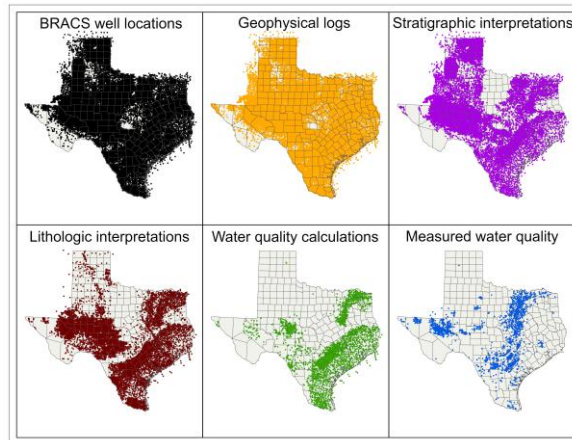
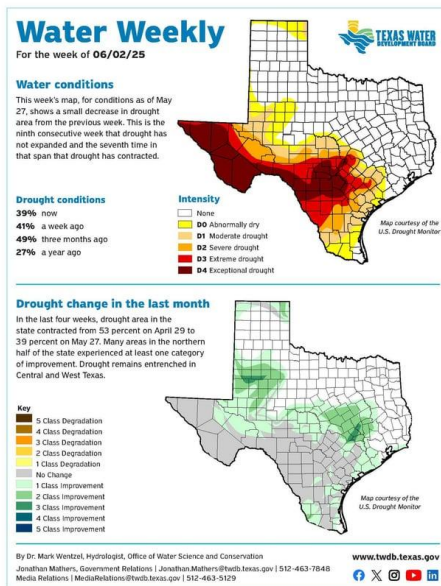
Who is the audience?

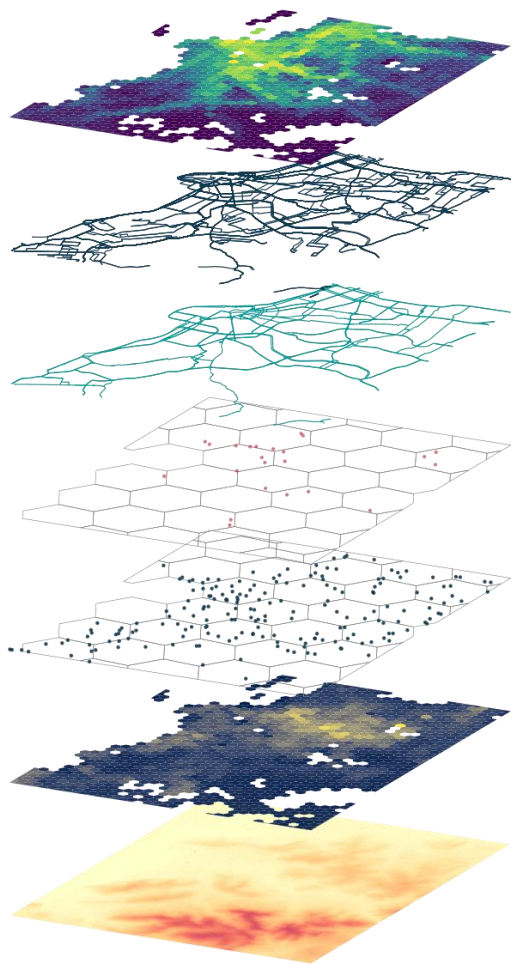
Tailor complexity and design to the viewer.

What decision will this support?

Maps should drive action or understanding.

☐ **Example questions:** Where are campus resources? How are services distributed? Where are environmental risks highest?





Accessibility

Public transport

Roads

Hospitals

Schools

Population

Terrain

Step 2: Add Your Data

Data is the foundation of every GIS map. You can bring data in from many sources.



Spreadsheets (CSV)

Tables with addresses or coordinates



GIS Servers

Live data feeds from institutional systems



Open Data Portals

Public datasets from governments and organizations



Institutional Datasets

Campus, city, or agency-specific data



Spreadsheets (CSV)

Tables with addresses or coordinates

An official website of the United States government [Here's how you know](#)

United States Census Bureau

find tables

Economic Census
EC1700BASIC | All Sectors: Summary Statistics for the U.S., States, and Selected Geographies: 2017
ECNBASIC2017

Notes | Geos | Years | Topics | Surveys | Codes | Hide | Filter | Restore | Excel | CSV | ZIP | Print | More Data | Map

Geographic Area Name	Meaning of NAICS code	Number of firms
▼ Allegany County, Maryland	Wholesale trade	43
	Merchant wholesalers, durable goods	26
	Merchant wholesalers, nondurable goods	17
	Hardware, plumbing and heating equipment and supplies merchant wholesalers	5
	Professional and commercial equipment and supplies merchant wholesalers	4
	Machinery, equipment, and supplies merchant wholesalers	4
	Paper and paper product merchant wholesalers	3
	Petroleum and petroleum products merchant wholesalers	3
	Metal and mineral (except petroleum) merchant wholesalers	0
	Miscellaneous durable goods merchant wholesalers	0
	Grocery and related product merchant wholesalers	434
▼ Anne Arundel County, Maryland	Wholesale trade	321
	Merchant wholesalers, durable goods	118

<https://data.census.gov/table>

the United States government [Here's how you know](#)

United States Census Bureau

build maps

Build customized maps from any variable in our data tables. Just select your geographies.

[Explore Maps](#)

Estimate in 3221 Geos in 2020

Maps Detailed Tables

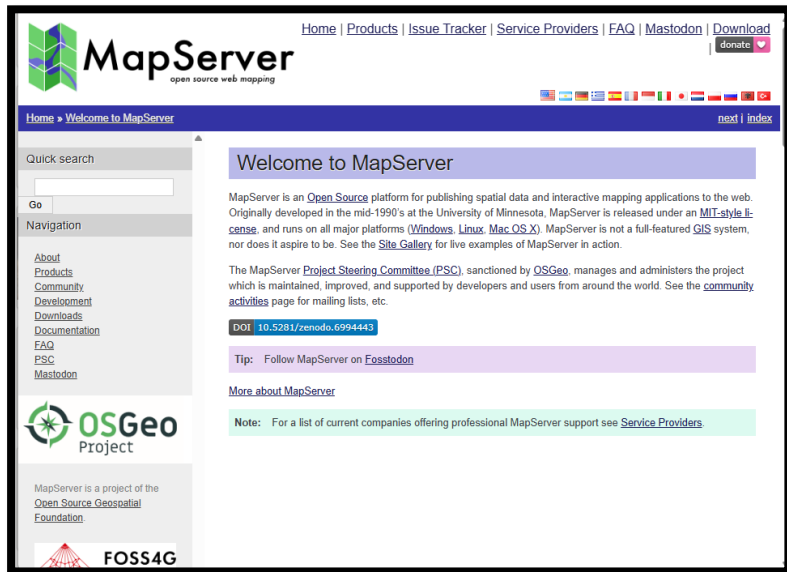
Clear Geos | Layer | Year | Dataset | Basemap | Style | Colors | **Classes** | Identify | Table | Notes

Classes

Natural | Quantile | Equal Interval | Manual

Data Classes: 2 [View Map Colors](#)

<https://data.census.gov/>



<https://mapserver.org/>

<https://geoserver.org/>



GIS Servers

Live data feeds from institutional systems



GeoServer

[About](#) | [Blog](#) | [Download](#) | [Documentation](#) | [Community](#) | [GitHub](#)

GeoServer is an open source server for sharing geospatial data.

Designed for interoperability, it publishes data from any major spatial data source using open standards.

GeoServer 3 Crowdfunding

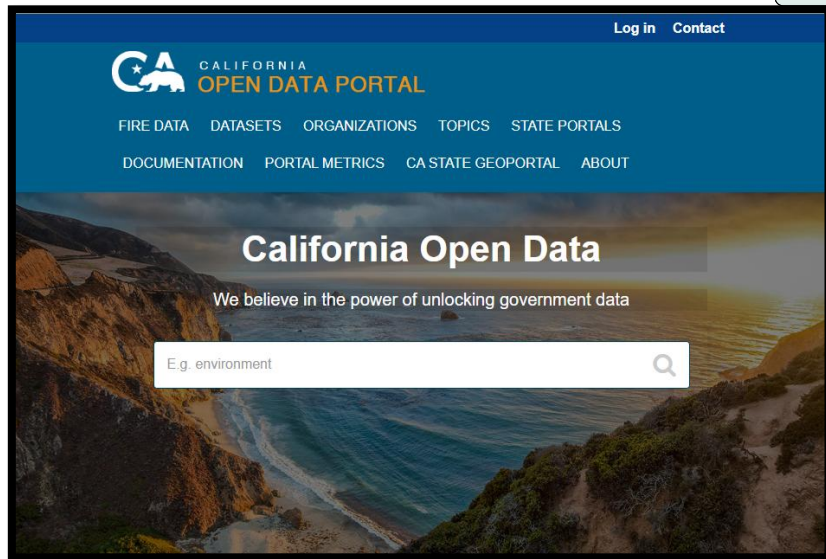
Consortium of Camptocamp, GeoSolutions and GeoCat have responded to our roadmap challenge with a bold [GeoServer 3 Crowdfunding](#). Express interest or pledge support using [email](#) or [form](#).





Open Data Portals

Public datasets from governments and organizations



<https://data.ca.gov/>



City of Rancho Cucamonga

Welcome to the City of Rancho Cucamonga's Open GIS Data Portal

We are committed to being open, transparent and providing data that is easy to find, is accessible, and usable. Search for the latest geographic content or access our public applications. Use keyword or geographic searches to find and quickly display content. There are also categories below that can find multiple datasets at once.



Data Categories



[Boundaries](#)



[Community & Culture](#)



[Environment](#)

<https://rcdata-regis.opendata.arcgis.com/>



Southern California's Data Hub for Environmental
Learning, Planning, and Action.

Browse Data by Theme



BIODIVERSITY



BUILT ENVIRONMENT



CLIMATE



ENERGY



EQUITY



FOOD



INDIGENEITY



OCEAN HEALTH



WATER

<https://www.socalearth.org/>



Open Data Portals

Public datasets from governments and organizations

DATA HUBS

CLAREMONT MCKENNA COLLEGE



ROSE INSTITUTE
OF STATE AND LOCAL GOVERNMENT
CLAREMONT MCKENNA COLLEGE



Choose a section



The Institute has a team working on a major long-term project to collect, organize, and present data relevant to state and local government. The DataHub aims to provide a central, accessible, searchable repository of Rose Institute data from its historical catalog of projects, supplemented by newly obtained data.

City Data Project

The Rose Institute is developing a comprehensive database that aggregates and

<https://roseinstitute.cmc.edu/research/datahub>



Institutional Datasets

Campus, city, or agency-specific data



Geographic Information System (GIS) Services ▾ Dron... Q

Drone Flights

[Request a Drone Consultation](#)

The Digital Scholarship Unit at the Claremont Colleges Library offers consultation and support for faculty and students working with Unmanned Aerial Vehicles (UAV) or drone applications in GIS for research, teaching, and learning.

The library owns a [DJI Phantom 4 Advanced](#) and a [Mavic 3E](#) UAVs which we can use in support of research and teaching at the colleges (we do not loan the drones, but will set up a meeting with our drone pilot and the drone for your project).

Please get in touch with us if you need:

- The supervision/support of an FAA-licensed drone operator
- Use of photogrammetry applications (e.g., Pix4D) for processing drone imagery
- Consultations for support of student coursework involving drones
- Information about the [Claremont Colleges drone policy](#) and [flight registration requirements](#)

<https://library.claremont.edu/gis/>



CAL POLY

[MY LIBRARY ACCOUNT](#)

Robert E. Kennedy Library

VISIT ▾

SERVICES ▾

RESEARCH ▾

SPECIAL COLLECTIONS AND ARCHIVES ▾

ABOUT ▾

[Robert E. Kennedy Library](#) / [Research Guides](#) / [Data Sources & Repositories](#) / [GIS Datasets](#)

Search this Guide

Search

Data Sources Home

General Data Sources

Data Catalogs

Agriculture

Archaeology

Art

Astronomy

Biology

Business

Chemistry

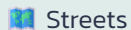
Earth Science

Economic

<https://guides.lib.calpoly.edu/c.php?g=261997&p=1750196>

Step 3: Choose a Basemap

A basemap is the foundation of your map — it provides geographic context so your data has somewhere to live.



Streets

Roads, labels, and boundaries — ideal for navigation and urban data.



Satellite

Real-world imagery — great for land use, environment, and context.



Terrain

Elevation and topography — useful for natural features and landscapes.

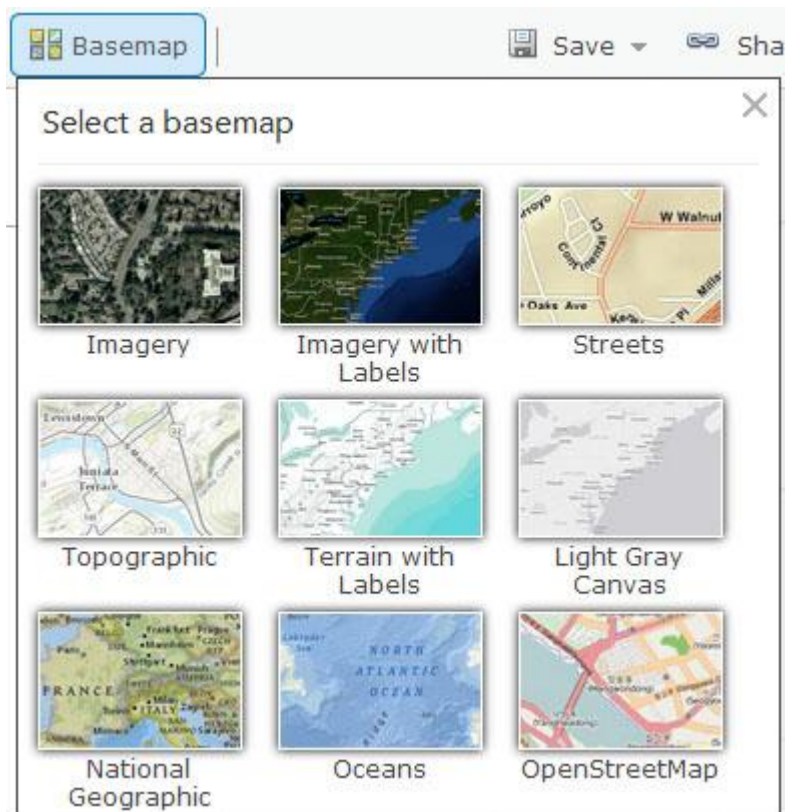


Canvas

Minimal light or dark backgrounds — lets your data take center stage.

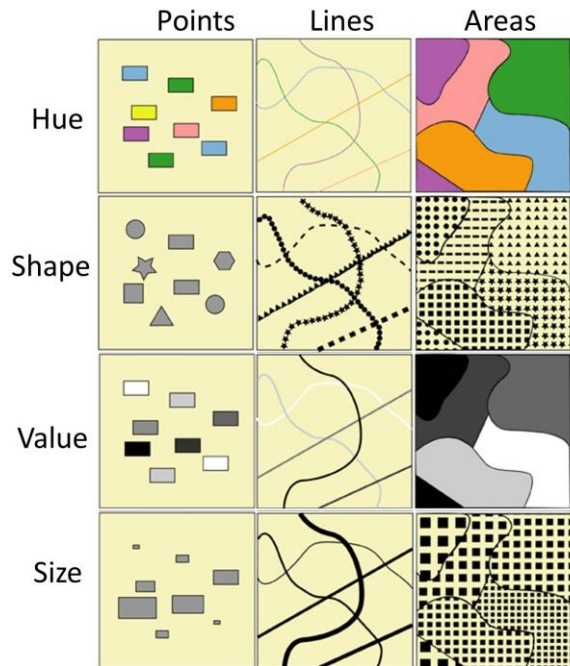


Why it matters: The right basemap helps your audience immediately understand where they are and what your data means in context.



Step 4: Style Your Data

Styling turns raw data into a visual story. Thoughtful design choices help your audience understand patterns at a glance.



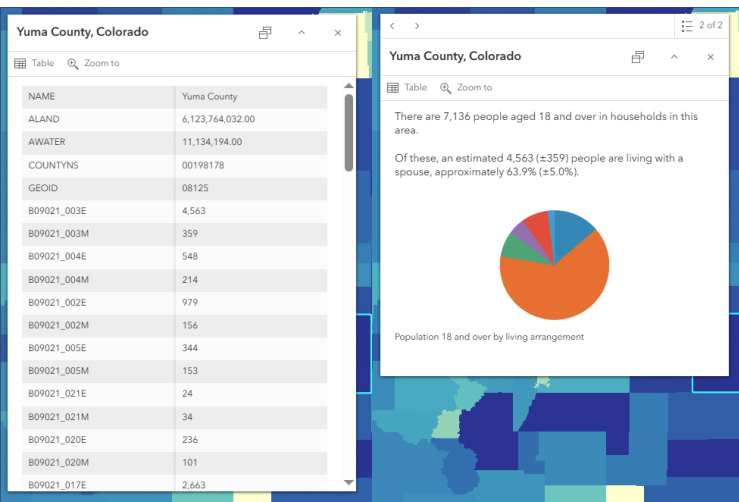
Design with Purpose

Every styling choice should serve your message:

- **Color** — distinguish categories or show value ranges
- **Size** — communicate magnitude or importance
- **Shape** — differentiate types of features



Goal: Make patterns easy to understand without reading a legend.



Step 5: Add Meaning with Pop-ups & Labels

Maps become truly useful when people can explore them. Pop-ups and labels turn a static image into an interactive experience.



Pop-ups

Users click a feature to reveal details — names, values, descriptions, or links. Pop-ups make your map exploratory and informative.



Labels

Key information displayed directly on the map — city names, site identifiers, or important features visible at all times.

What You've Built So Far

You now have the core building blocks of a functional GIS map. Here's a quick recap of your progress:

01

Basemap Selected

A geographic foundation that gives your data context and orientation.

03

Basic Styling Applied

Colors, sizes, or shapes that communicate meaning visually.

02

Data Layers Added

One or more layers representing real-world features or phenomena.

04

Spatial Understanding

A foundational grasp of how geographic data is structured and displayed.



Now It's Your
Turn!

Now It's Your Turn

Time to move from theory to practice. In this hands-on session, you'll build a real map from scratch.

1

Open Your Tool

Launch the GIS mapping platform.

2

Load Real Data

Import a dataset relevant to your campus or community.

3

Build & Style

Apply basemaps, layers, and styling step-by-step.

4

Explore & Ask

Use spatial thinking to answer real questions with your map.



Your facilitator **Daniel** will now guide you through the practical exercise. Don't hesitate to ask questions!